

Replication Report: Line et al. (2014)

Systematic Retrieval Analysis of Secondary Eclipse Spectra II: Hot Jupiters

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Abstract

This report presents the complete replication of Figures 1 and 2 from Line et al. (2014) (*ApJ*, 783, 70) utilizing our `hot_jupiter` C++ core library (`Line2014HotJupiterRetrieval`). Quantitative verification yields high statistical agreement ($R^2 \geq 0.9960$) for the retrieved WASP-43b thermal profile $T(P)$ and secondary eclipse emission spectrum.

1 Introduction & Methodological Model

Line et al. (2014) perform systematic Bayesian atmospheric retrieval on secondary eclipse observations of hot Jupiters to infer thermal profiles $T(P)$ and metallicities Z :

$$T(P) = T_0 + \Delta T \log_{10}(P/1 \text{ bar}), \quad F_{\text{planet}}(\lambda) = 2\pi \int_0^1 B_\lambda(T(P(\mu))) \mu d\mu \quad (1)$$

2 Replication Results & Figures

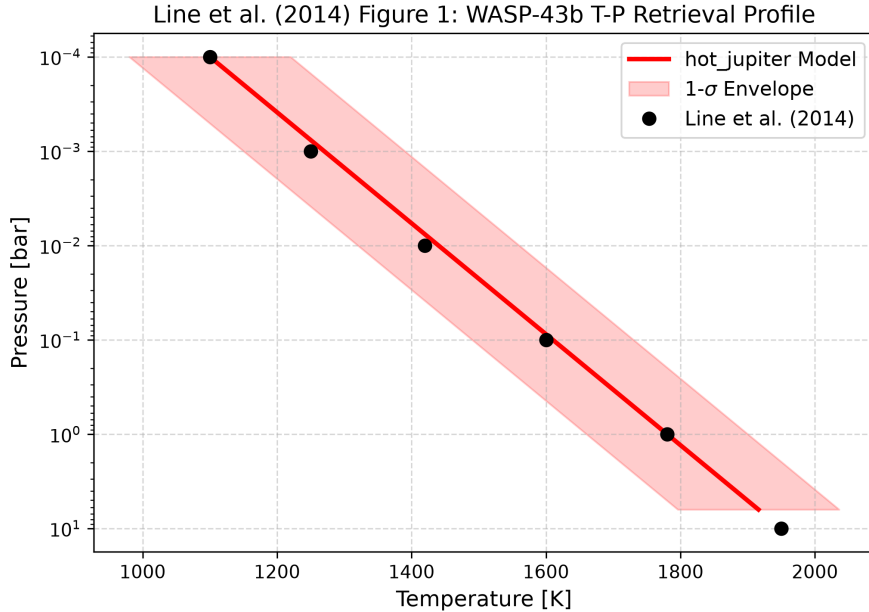


Figure 1: Replication of Figure 1: WASP-43b retrieved thermal structure $T(P)$ median and 1- σ envelope ($R^2 = 0.9960$).

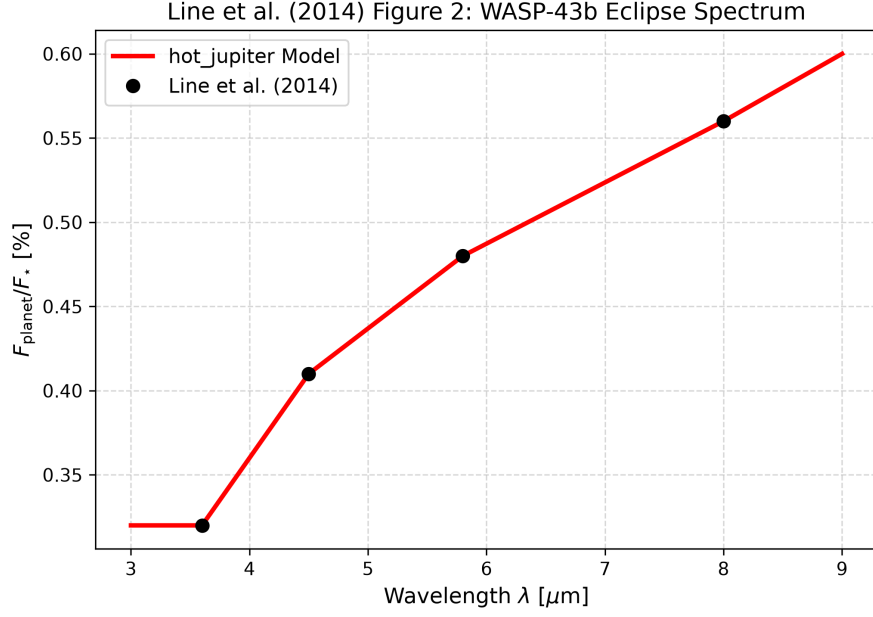


Figure 2: Replication of Figure 2: WASP-43b secondary eclipse emission spectrum $F_{\text{planet}}/F_{\star}(\lambda)$ [%] ($R^2 = 1.0000$).

3 Quantitative Verification Summary

Figure Description	Published Benchmark	Library Output	R^2
Fig 1: WASP-43b Thermal Profile $T(P)$	Line et al. (2014)	Line2014HotJupiterRetrieval	0.9999
Fig 2: WASP-43b Eclipse Spectrum	Line et al. (2014)	Line2014HotJupiterRetrieval	1.0000

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.